Gurumauj Satsangi

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Portfolio | GitHub | LinkedIn

Profile

I am a final-year B.Tech Computer Science and Engineering student at Vellore Institute of Technology (VIT), India. I build applied machine learning and AI systems, mostly real-time and agentic pipelines, and I have three accepted conference papers on this work. I am interested in how people work with AI systems, especially how AI assistance affects human understanding and confidence, and I want to learn how to design and analyse human-subject studies.

Education

Bachelor of Technology in Computer Science and Engineering, Vellore Institute of Technology, Vellore, India, 2023 to 2027 (expected).

CGPA: 8.43 out of 10

Publications

1. G. Satsangi, K. Maheshwari, Narayanan Prasanth N "**LiveLingo: An End-to-End Streaming Middleware for Real-Time Translation of Livestreams into Indic Languages**" Accepted for oral presentation and publication in the proceedings of IEEE InCODE 2026, Oriental Institute of Science & Technology, Bhopal
2. V. Khurana, G. Satsangi, A. Sharma, and M. Umamaheswari. "**A System and Method for Voice-Driven Semantic Image Search Leveraging AWS and OpenAI**" Accepted for oral presentation and publication in the proceedings of IEEE INCECT 2026, Jain College of Engineering, Belagavi *(IEEE Xplore)*
3. G. Satsangi, V. Khurana, A. Sharma, and M. Umamaheswari "**Agentic Multi-Source Validation of Healthcare Provider Directories Using Workflow Orchestration and Confidence-Based Decision Routing**" Accepted for poster presentation at the 49th (Inter)National Systems Conference (NSC 2026), which was held jointly with the The 8th International Conference on Dayalbagh (Art) Science (& Engineering) of (Evolutionary/Re-Evolutionary) Consciousness (DSC) organised by a consortium of seven universities, namely the Dayalbagh Educational Institute, Deemed to be University (DEI) in India, Kiel University (CAU) in Germany, the University of Birmingham, UK, the University of Waterloo in Canada, Western University in Ontario, Canada, the University of British Columbia in Vancouver, Canada, and the Arizona State University, USA & the Systems Society of India

Research-related projects

RASHMI (Retrieval & AI-based Smart Healthcare Medical Imaging) is an IoT-enabled optical X-ray digitizer with cloud-based semantic retrieval for diagnostic support. An ESP32 push button triggers a USB camera in a back-lit, battery-backed imaging chamber; a Node.js pipeline crops each film with OpenCV, embeds it with DenseNet-121 (PadChest weights, served through ONNX Runtime) and a Rad-DINO ensemble, retrieves similar reported cases from Qdrant, and generates a summary with GPT-4o. *(Ongoing Project)*

LiveLingo is a system for real-time translation of livestreams into Indic languages. I built a WebSocket pipeline that connects AWS IVS to a translation service and reaches under 400 ms end to end on a constrained 350 Kbps network, including middleware that absorbs TCP head-of-line blocking spikes of up to 17.3 seconds. It is the basis of my InCODE paper. *(GitHub)*

Validex is a multi-agent healthcare provider data validation platform. It uses LangGraph workflow orchestration and confidence-based routing, processes 100 provider records in under 5 minutes with over 85% extraction accuracy from unstructured PDFs, and cuts manual verification work by 80 to 90%. It was selected for the semi-finals of EY Techathon 6.0 (2026) and is the subject of my NSC poster. (GitHub / Live)

Memory Lens is a voice-driven semantic image search system built with Flutter, AWS Rekognition, OpenAI embeddings, and Qdrant. It answers queries in under 2 seconds across more than 1,000 images with over 85% semantic relevance and over 95% voice recognition accuracy. It reached the semi-finals of the Samsung PRISM GenAI Hackathon (2025) and is the basis of my INCECT paper. *(GitHub / Demo Video)*

Experience

SDE Intern, Algoleap Technologies, Hyderabad (May to July 2025). I designed a single GenAI API covering text, image, audio, and video generation, which removed the need for four separate API keys and cut integration and rate-limit errors by 30%.

Intern, Indian Association for the Cultivation of Science, Kolkata (December 2024 to January 2025). I built a login-secured Store Inventory and Job Card Portal that improved departmental efficiency by 40% and handled more than 8,000 applications with no downtime.

Summer Intern, Dayalbagh Educational Institute, Agra (June to July 2024). I developed a Moodle-based online learning platform for the ICT Continuing Education Centre, supporting more than 100 learners.

Honours

- Honourable Mention, Bangkok University x Berkeley SkyDeck Fund Hackathon (2026).
- Best AI Model, Yantra Central Hackathon, VIT (2025).
- Winner, Aquathon, VIT (2024).
- Semi-finalist, EY Techathon 6.0 (2026) and Samsung PRISM GenAI Hackathon (2025).

Skills and certifications

- Programming in Python, Java, JavaScript, and SQL.
- Backend work in Node.js and Express.
- AI tools including LangGraph, the OpenAI API, and prompt engineering.
- Deep Learning & Medical Imaging: DenseNet-121, Rad-DINO, embedding-based image retrieval, OpenCV
- Cloud and DevOps with AWS, Docker, and CI/CD.
- Databases: MySQL, PostgreSQL, Redis, Neo4j, and MongoDB.
- Certifications: AWS Certified Cloud Practitioner (January 2026), Solutions Architect Associate (June 2026), and Developer Associate (June 2026).